Solving a generalisation of tic-tac-toe or k-in-a-row game on finite and infinite $\{m, n\}$ tilings of 2D space

Subtitle

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1 Introduction

1.1 Notation

1.1.1 Schläfli symbol

The Schläfli symbol is used to compactly denote any regular polytope or tessellation. For our case it will be of the form $\{p, q\}$, where $\{p\}$ is a regular p-sided polygon and q is the number of these polygons joined at each vertex.

1.1.2 mnk-notation

To denote a game variant played on a $\{m, n\}$ tiling with the winning objective of getting k in a row, we will use the notation $\{m, n\}^k$. For example, the classical tic-tac-toe game is denoted as $\{4, 4\}^3$.

1.1.3 Game versions

$(1, 1, 1)$ where each player occupies 1 position in a move. $(1, 2, 2)$ where the first move occupies 1 position and then both players occupy 2 positions per move.

1.2 Maker-Breaker games

In a Maker-Breaker game, the Maker is the player trying to achieve the winning objective and Breaker is the player trying to stop the Maker from achieving this.

1.2.1 Insights into Maker-Maker games

If a k-game is a Breaker win in the Maker-Breaker version, we know that the original Maker-Maker version is a draw. Conversely if a k-game is a Maker win in the Maker-Breaker version, we cannot tell anything about the original Maker-Maker variant. Therefore to prove Maker wins, we have to show them in the Maker-Maker version.

1.2.2 Strategy stealing argument for $(1, 1, 1)$

In a Maker-Maker game, if there was a strategy such that P2 would have a winning strategy, P1 could just steal the strategy in a way where P1 would make a random move and then effectively play the role of P2 and employ that winning strategy. This is a contradiction as both P1 and P2 have a winning strategy. Therefore only P1 can have a winning strategy.

1.2.3 Pairing strategies

A pairing strategy is a strategy for Breaker where Breaker pairs up positions on the board in such a way that if Maker occupies one position in a pair, Breaker can immediately occupy the other position in the pair and thus prevent Maker from winning. If Breaker can successfully create such a pairing strategy that covers all possible winning combinations for Maker, then Breaker has a winning strategy.

1.3 Insights

In $\{m, n\}$ tiling let

$$m = 2i : i \in \mathbb{N}. \tag{1}$$

It matters so that there ale straight lines going through the shapes. Then in $n = 2j + 1$ tilings, diagonals do not make sense. In $n = 2j$ tilings, however, diagonals make sense. We then distinguish between a $\{m, n = 2j\}_m$ and $\{m, n = 2j\}_{vn}$ configurations named after the Moore (M) and Von Neumann (VN) neighbourhoods. The Moore version permitting diagonals to be used and the Von Neumann version only permitting edge-connected series. The classical tic-tac-toe is then played on a $\{4, 4\}_m$ board. In the spirit of m, n, k games we include also the the length of the winning sequence as follows: $\{m, n\}^k$. The number of axis that you can win on then becomes m in the M configuration and $m/2$ in the VN configuration.

2 Tiling spaces

No spherical tilings since they are all finite and do not form an infinite playing board.

The proof for any tiling with $k = 1$ is trivial as the first player wins by making any move.

m \n	**2**	**3**	**4**	**5**	**6**	**7**	**8**	∞
2	$\{2,2\}$ Spherical	$\{2,3\}$ Spherical	$\{2,4\}$ Spherical	$\{2,5\}$ Spherical	$\{2,6\}$ Spherical	$\{2,7\}$ Spherical	$\{2,8\}$ Spherical	$\{2,\infty\}$ Euclidean
3	$\{3,2\}$ Spherical	$\{3,3\}$ Platonic	$\{3,4\}$ Platonic	$\{3,5\}$ Platonic	$\{3,6\}$ Euclidean	$\{3,7\}$ Hyperbolic	$\{3,8\}$ Hyperbolic	$\{3,\infty\}$ Paracompact
4	$\{4,2\}$ Spherical	$\{4,3\}$ Platonic	$\{4,4\}$ Euclidean	$\{4,5\}$ Hyperbolic	$\{4,6\}$ Hyperbolic	$\{4,7\}$ Hyperbolic	$\{4,8\}$ Hyperbolic	$\{4,\infty\}$ Paracompact
5	$\{5,2\}$ Spherical	$\{5,3\}$ Platonic	$\{5,4\}$ Hyperbolic	$\{5,5\}$ Hyperbolic	$\{5,6\}$ Hyperbolic	$\{5,7\}$ Hyperbolic	$\{5,8\}$ Hyperbolic	$\{5,\infty\}$ Paracompact
6	$\{6,2\}$ Spherical	$\{6,3\}$ Euclidean	$\{6,4\}$ Hyperbolic	$\{6,5\}$ Hyperbolic	$\{6,6\}$ Hyperbolic	$\{6,7\}$ Hyperbolic	$\{6,8\}$ Hyperbolic	$\{6,\infty\}$ Paracompact
7	$\{7,2\}$ Spherical	$\{7,3\}$ Hyperbolic	$\{7,4\}$ Hyperbolic	$\{7,5\}$ Hyperbolic	$\{7,6\}$ Hyperbolic	$\{7,7\}$ Hyperbolic	$\{7,8\}$ Hyperbolic	$\{7,\infty\}$ Paracompact
8	$\{8,2\}$ Spherical	$\{8,3\}$ Hyperbolic	$\{8,4\}$ Hyperbolic	$\{8,5\}$ Hyperbolic	$\{8,6\}$ Hyperbolic	$\{8,7\}$ Hyperbolic	$\{8,8\}$ Hyperbolic	$\{8,\infty\}$ Paracompact
∞	$\{\infty,2\}$ Euclidean	$\{\infty,3\}$ Paracompact	$\{\infty,4\}$ Paracompact	$\{\infty,5\}$ Paracompact	$\{\infty,6\}$ Paracompact	$\{\infty,7\}$ Paracompact	$\{\infty,8\}$ Paracompact	$\{\infty,\infty\}$ Paracompact

2.1 Spherical tilings

All spherical tilings are finite and therefore do not provide an infinite playing board.

2.1.1 The $\{m, 2\}$ tilings

The $\{m, 2\}$ tilings consist of only 2 playable polygons, and therefore the game is a trivial draw for $k > 1$ as both of the players occupy exactly one of the polygons.

2.1.2 The $\{2, n\}$ tilings

Proof is the same as for the $\{2, \infty\}$ tiling.

2.2 Platonic tilings

Platonic tilings are a subset of phericla tilings that result in the platonic solids. They are the $\{3, 3\}, \{3, 4\}, \{3, 5\}, \{4, 3\}, \{5, 3\}$ tilings and correspond to the tetrahedron, octahedron, icosahedron, cube and dodecahedron respectively. Out of these we will only look at $\{4, 3\}$ as all of the other tilings are made of odd-sided polygons and therefore do not satisfy condition 1.

2.2.1 The $\{4, 3\}$ tiling

This platonic tiling corresponds to the cube. It is a trivial draw for $k > 3$ as both players alternatively occupy one of the polygons and since there are 6 of them, each player will in the end occupy 3 of the polygons.
For $K \leq 3$ it is a win for P1. This can be shown by brute force as there are only 8 possible games (up to symmetry) for the $\{4, 3\}^3$ game.

2.3 Euclidean tilings

The only regular tilings of Euclidean space are $\{2,\infty\},\{3,6\},\{4,4\},\{6,3\},\{\infty,2\}$. We do not look at the $\{\infty,2\}$ tiling as it does not provide an infinite playing board. We also exclude the $\{3,6\}$ tiling as it does not satisfy condition 1. Finally, we also do not spend too much time looking at the $\{4,4\}$ tiling as this is the case with regular m,n,k-games and has been studied extensively. We are left with the $\{2,\infty\}$ and $\{6,3\}$ tilings, which we take a close look at.

2.3.1 The $\{4,4\}$ tiling

This tiling corresponds to the regular square tiling of the classical tic-tac-toe game. It has been shown to be a draw for $k \geq 8$ by Zetters [9] and a win for P1 for $k \leq 5$ by Victor ALlis [1].

2.3.2 The $\{2,\infty\}$ tiling

This is a very trivial case as it is a win for P1 in the case of $k \leq 2$ and a draw in the case $k > 2$.

For the case $k = 2$ P1 has a winning strategy. We can see that the $\{2,\infty\}$ has a reflective symmetry. Therefore the is only one game (up to symmetry) of $\{2,\infty\}^2$. It is shown in Figure 1.

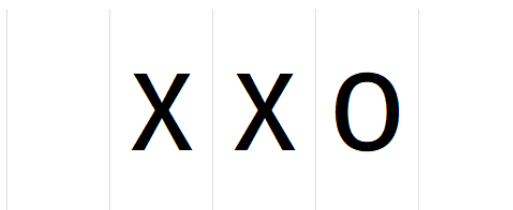

Figure 1: Drawing of the winning strategy for $\{2,\infty\}^2$.

For the case $k > 2$ We can show this using a simple pairing argument where we tile the space as in Figure 2.

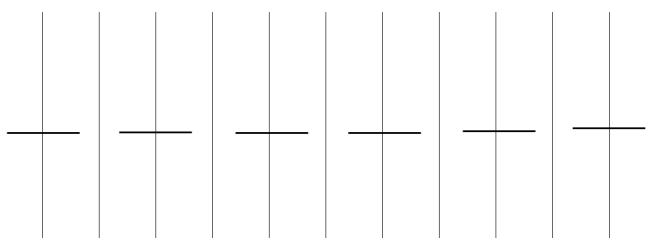

Figure 2: Pairing strategy for $\{2,\infty\}^3$.

NEEDS PROOF OF SAME THING FOR $(1,2,2)$

2.3.3 The $\{6,3\}$ tiling

Since the vertex degree in this tiling is odd, we will not consider diagonals.

Figure 3 shows a winning breaker strategy for the $\{6,3\}^7$ game. The lines represent pairs like in a regular pairing strategy. The rest is tiled by the 4x3 rhombuses. A zoomed in version of one of these tiles is in Figure 4. The couloured lines represent all the directions in which breaker has to play at least once to win.
This has been shown to be a draw by T. G. L. Zetters [9].
NEEDS PROOF FOR $k < 5$**.**

NEEDS PROOFS FOR $(1,2,2)$

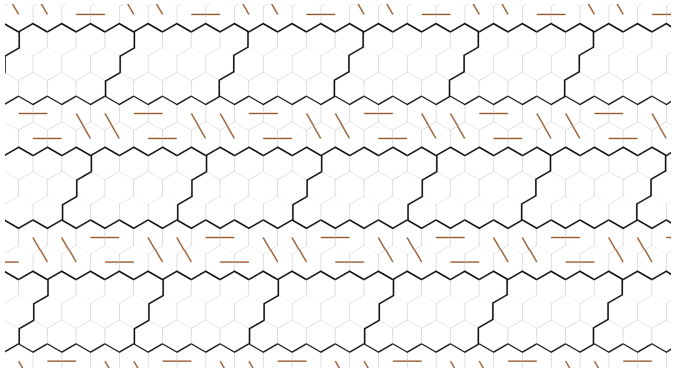

Figure 3: Pairing strategy for $\{6,3\}^7$.

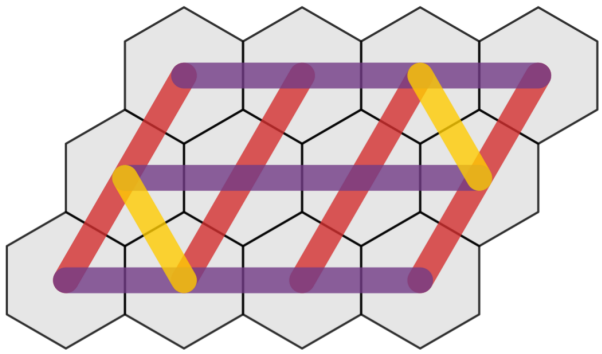

Figure 4: One repeating tile in the $\{6,3\}^7$ pairing strategy.

2.4 Hyperbolic tilings

There is an infinite amount of hyperbolic tilings. We will therefore look at some representative examples and then draw general conclusions.

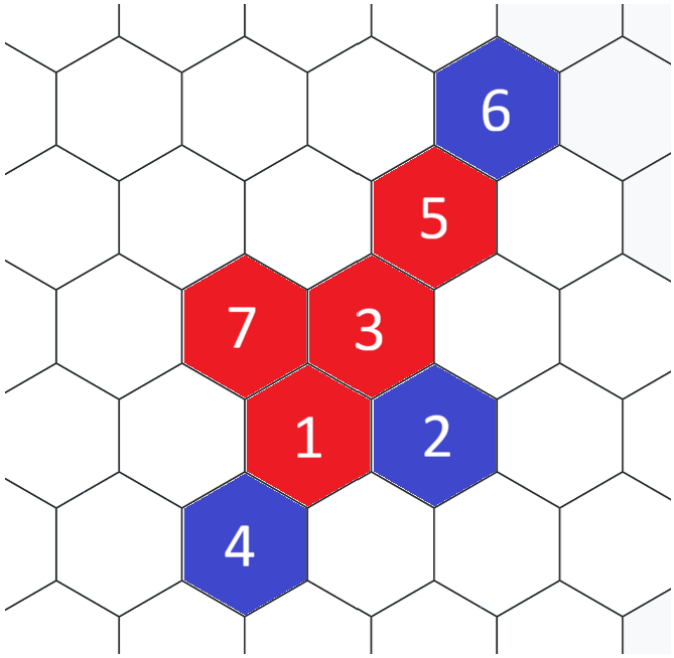

Figure 5: Forced Winning configuration (up to symmetry) for $\{6, 3\}^4$.

References

[1] L. Victor Allis. *Searching for Solutions in Games and Artificial Intelligence*. PhD thesis, University of Limburg, Maastricht, Netherlands, 1994. Available at `http://fragrieu.free.fr/SearchingForSolutions.pdf`.

[2] L. Victor Allis, H. Jaap van den Herik, and Jos W. H. M. Uiterwijk. Go-moku and threat-space search. In *Proceedings of the 13th International Joint Conference on Artificial Intelligence (IJCAI-93)*, pages 225–230, Chambéry, France, 1993. Morgan Kaufmann.

[3] József Beck. Van der waerden and ramsey type games. 1(2):103–116, 1981.

[4] József Beck. *Combinatorial Games: Tic-Tac-Toe Theory*. Cambridge University Press, 2008.

[5] Elwyn R. Berlekamp, John H. Conway, and Richard K. Guy. *Winning Ways for Your Mathematical Plays, Volume 3*. A K Peters, 2 edition, 2003.

[6] Paul Erdős. Combinatorial problems and games. 1978.

[7] Paul Erdős and John L. Selfridge. On a combinatorial game. 12(6):876–881, 1961.

[8] Paul Erdős and John L. Selfridge. On a ramsey-type problem. 14(3):298–301, 1973.

[9] Richard K. Guy, J. L. Selfridge, and T. G. L. Zetters. S10. *The American Mathematical Monthly*, 87(7):575–576, 1980.

[10] Alfred W. Hales and Robert I. Jewett. Regularity and positional games. 106(2):222–229, 1963.

[11] Dan Hefetz, Michael Krivelevich, Miloš Stojaković, and Tibor Szabó. Avoider–enforcer games. 15(1):R42, 2008.

[12] Tim Hogg. Antwerp game configuration tool, 2020. Online interactive tool.

[13] Nick Kruczek, Carl Sundberg, et al. Set matching: An enhancement of the hales–jewett pairing strategy. 2017.

[14] John Tromp. The game of Connect-4. Master's thesis, University of Amsterdam, Amsterdam, Netherlands, 1995. Available at `https://tromp.github.io/c4/connect4_thesis.pdf`.

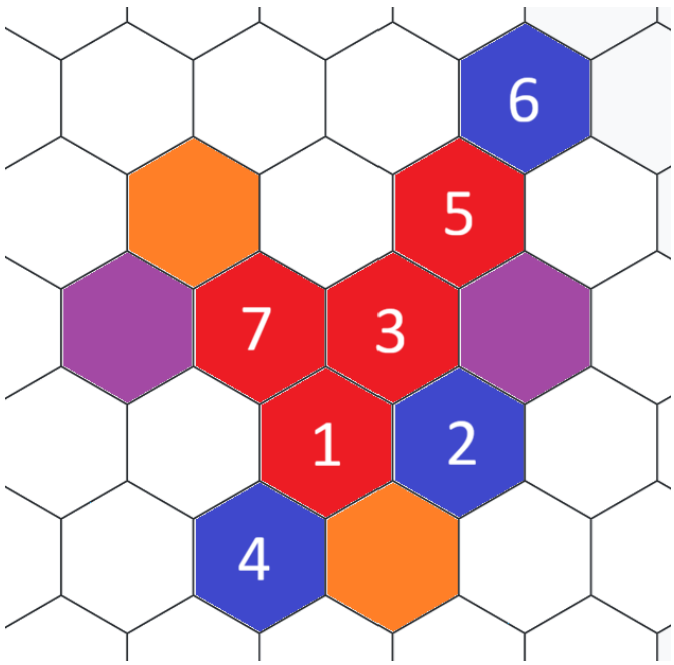

Figure 6: Threat illustration for $\{6, 3\}^4$.